

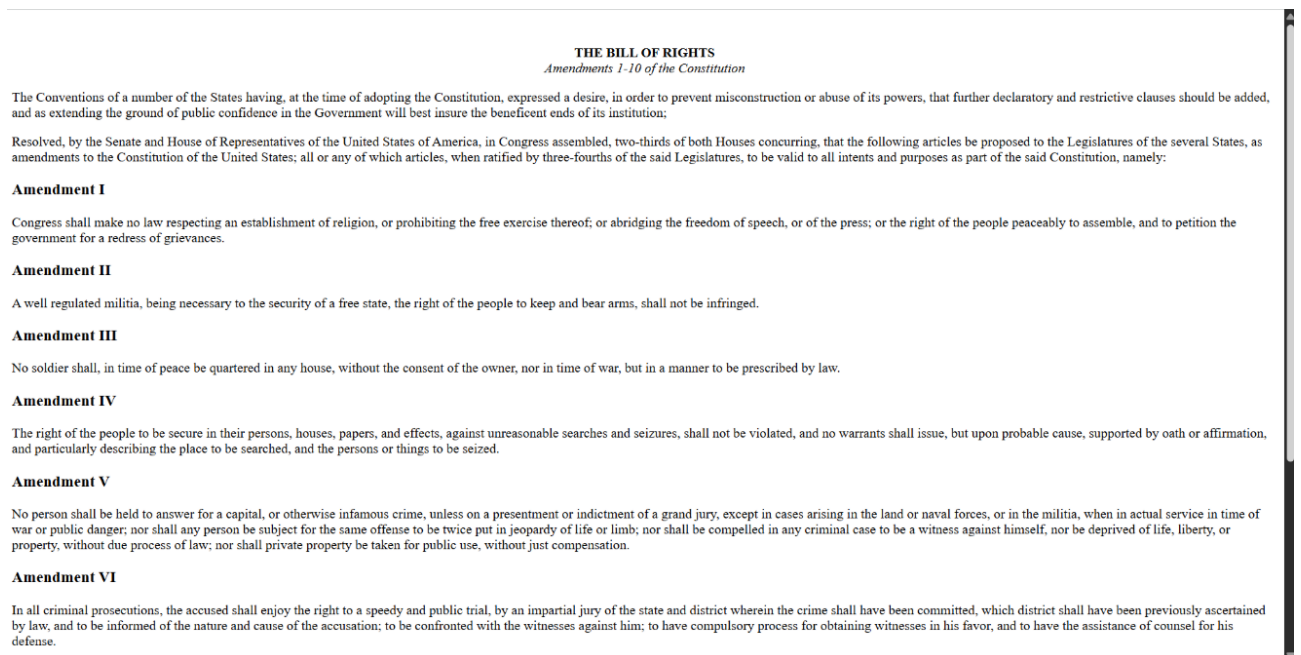
PENGUMPULAN TUGAS JARINGAN KOMPUTER Ethernet and ARP

Pengantar

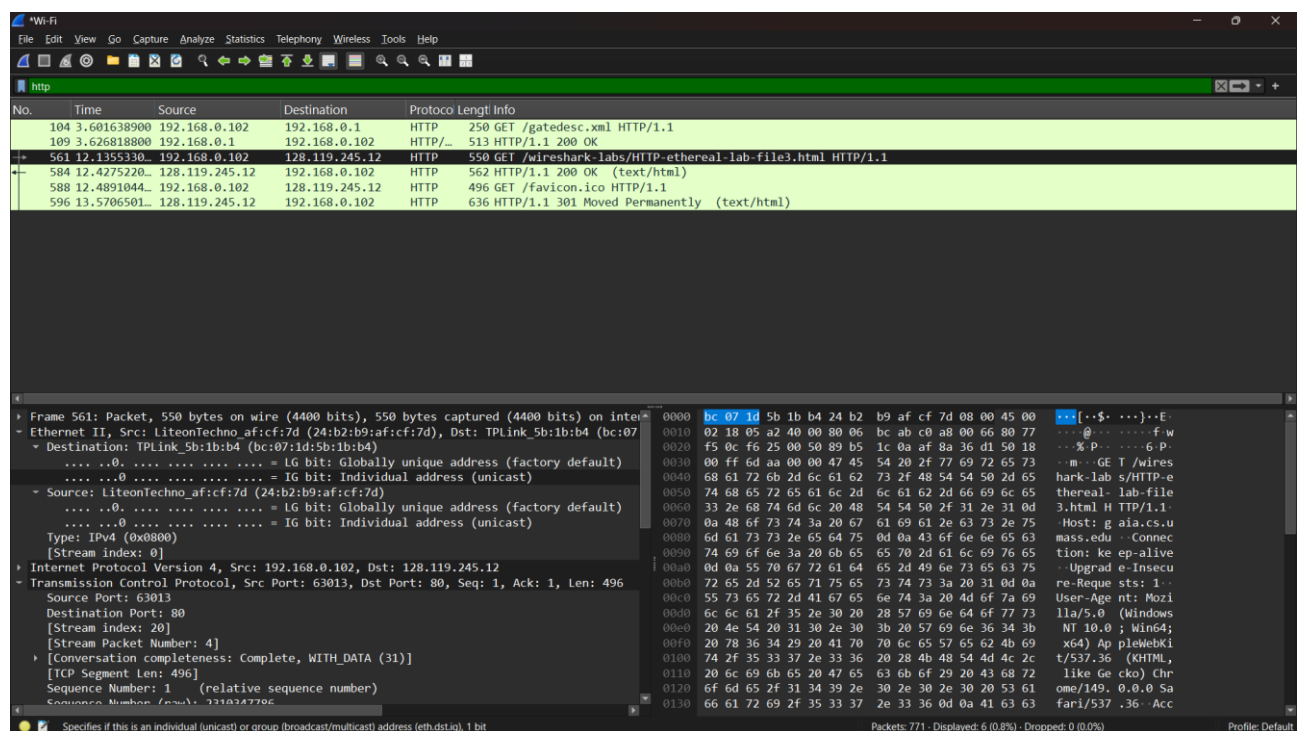
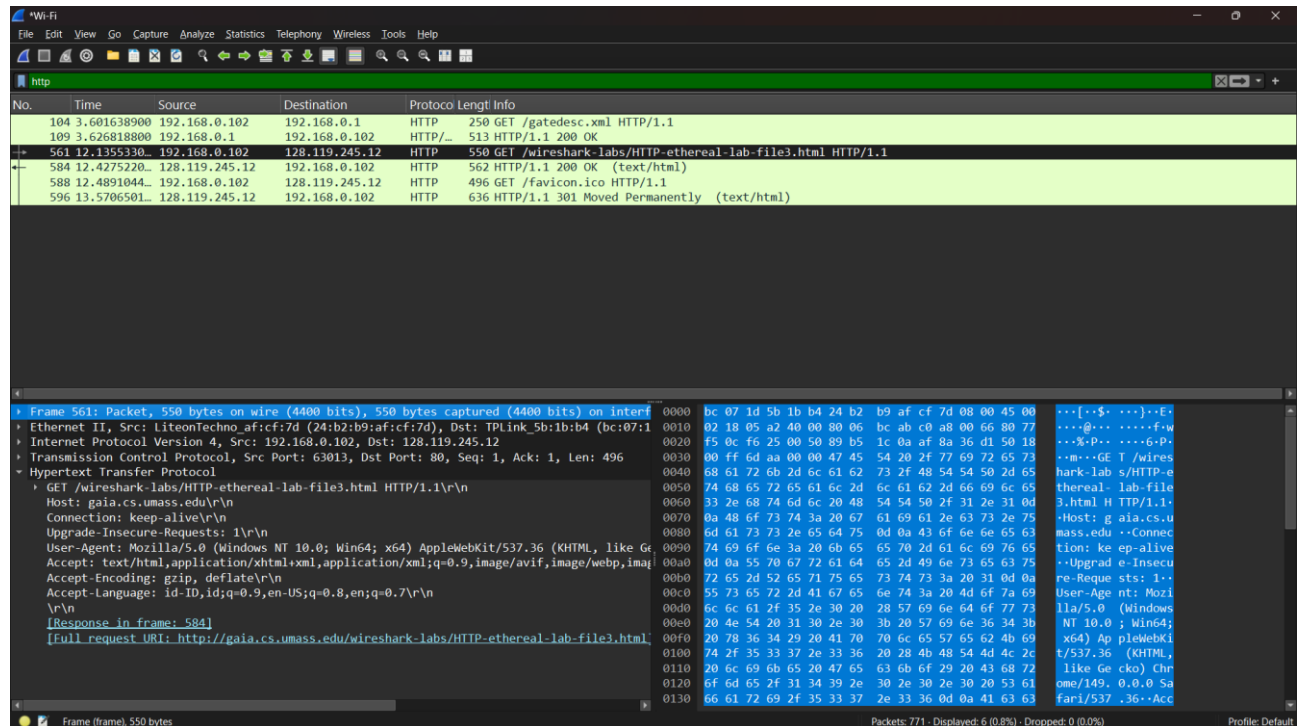
saya akan menyelidiki protokol Ethernet dan protokol ARP. RFC 826 ([ftp://ftp.rfc-editor.org/in-notes/std/std37.txt](http://ftp.rfc-editor.org/in-notes/std/std37.txt)) berisi detail mengenai protokol ARP, yang digunakan oleh perangkat IP untuk menentukan alamat IP dari antarmuka jarak jauh yang alamat Ethernetnya diketahui.

Menangkap dan menganalisis frame Ethernet

1. <http://gaia.cs.umass.edu/wireshark-labs/HTTP-ethereal-lab-file3.html>



2. HTTP after open gaia.cs.umass.edu/wireshark-labs/HTTP-ethereal-lab-file3.html



Address Resolution Protocol

Caching ARP

1. CMD arp -a

```
C:\WINDOWS\system32\cmd. x + v
Microsoft Windows [Version 10.0.26200.8457]
(c) Microsoft Corporation. All rights reserved.

C:\Users\fairuuz>arp -a

Interface: 192.168.0.102 --- 0xb
Internet Address      Physical Address      Type
192.168.0.1           bc-07-1d-5b-1b-b4    dynamic
192.168.0.255         ff-ff-ff-ff-ff-ff    static
224.0.0.22            01-00-5e-00-00-16    static
224.0.0.251           01-00-5e-00-00-fb    static
224.0.0.252           01-00-5e-00-00-fc    static
239.255.255.250       01-00-5e-7f-ff-fa    static
255.255.255.255       ff-ff-ff-ff-ff-ff    static

C:\Users\fairuuz>
```

2. paket ARP

The image shows a Wireshark packet capture of ARP traffic. The packet list pane displays several ARP packets, including requests and responses. The packet details pane shows the structure of an ARP request packet, including the Ethernet II header, Internet Protocol (IP) header, and ARP header.

Packet List:

No.	Time	Source	Destination	Protocol	Length	Info
33	11.3525907...	TPLink_5b:1b:b4	LiteonTechno_af:...	ARP	42	Who has 192.168.0.102? Tell 192.168.0.1
34	11.3526305...	LiteonTechno_af:...	TPLink_5b:1b:b4	ARP	42	192.168.0.102 is at 24:b2:b9:af:cf:7d
212	30.6122030...	TPLink_5b:1b:b4	Broadcast	ARP	42	Who has 192.168.0.105? Tell 192.168.0.1
219	32.5578756...	TPLink_5b:1b:b4	Broadcast	ARP	42	Who has 192.168.0.105? Tell 192.168.0.1
250	43.4185193...	TPLink_5b:1b:b4	LiteonTechno_af:...	ARP	42	Who has 192.168.0.102? Tell 192.168.0.1
251	43.4185484...	LiteonTechno_af:...	TPLink_5b:1b:b4	ARP	42	192.168.0.102 is at 24:b2:b9:af:cf:7d
332	52.4241806...	TPLink_5b:1b:b4	Broadcast	ARP	42	Who has 192.168.0.105? Tell 192.168.0.1
333	53.4487294...	TPLink_5b:1b:b4	Broadcast	ARP	42	Who has 192.168.0.105? Tell 192.168.0.1
337	54.4734655...	TPLink_5b:1b:b4	Broadcast	ARP	42	Who has 192.168.0.105? Tell 192.168.0.1
500	94.7230723...	TPLink_5b:1b:b4	LiteonTechno_af:...	ARP	42	Who has 192.168.0.102? Tell 192.168.0.1
501	94.7231154...	LiteonTechno_af:...	TPLink_5b:1b:b4	ARP	42	192.168.0.102 is at 24:b2:b9:af:cf:7d
717	124.384280...	TPLink_5b:1b:b4	LiteonTechno_af:...	ARP	42	Who has 192.168.0.102? Tell 192.168.0.1
718	124.384306...	LiteonTechno_af:...	TPLink_5b:1b:b4	ARP	42	192.168.0.102 is at 24:b2:b9:af:cf:7d
845	164.257057...	TPLink_5b:1b:b4	LiteonTechno_af:...	ARP	42	Who has 192.168.0.102? Tell 192.168.0.1
846	164.257081...	LiteonTechno_af:...	TPLink_5b:1b:b4	ARP	42	192.168.0.102 is at 24:b2:b9:af:cf:7d
902	177.808469...	TPLink_5b:1b:b4	Broadcast	ARP	42	Who has 192.168.0.104? Tell 192.168.0.1
936	195.073700...	TPLink_5b:1b:b4	Broadcast	ARP	42	Who has 192.168.0.102? Tell 192.168.0.1
937	195.073718...	LiteonTechno_af:...	TPLink_5b:1b:b4	ARP	42	192.168.0.102 is at 24:b2:b9:af:cf:7d

Packet Details (Frame 33):

- Frame 33: Packet, 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface \Device\NPF{...}
- Ethernet II, Src: TPLink_5b:1b:b4 (bc:07:1d:5b:1b:b4), Dst: LiteonTechno_af:cf:7d (24:b2:b9:af:cf:7d)
- Address Resolution Protocol (request)

Packet Bytes:

Offset	Hex	ASCII
0000	24 b2 b9 af cf 7d bc 07	
0006	1d 5b 1b b4 08 06 00 01	
0010	08 00 06 04 00 01 bc 07	
0016	1d 5b 1b b4 c0 a8 00 01	
0020	00 00 00 00 00 c0 a8 00 66	